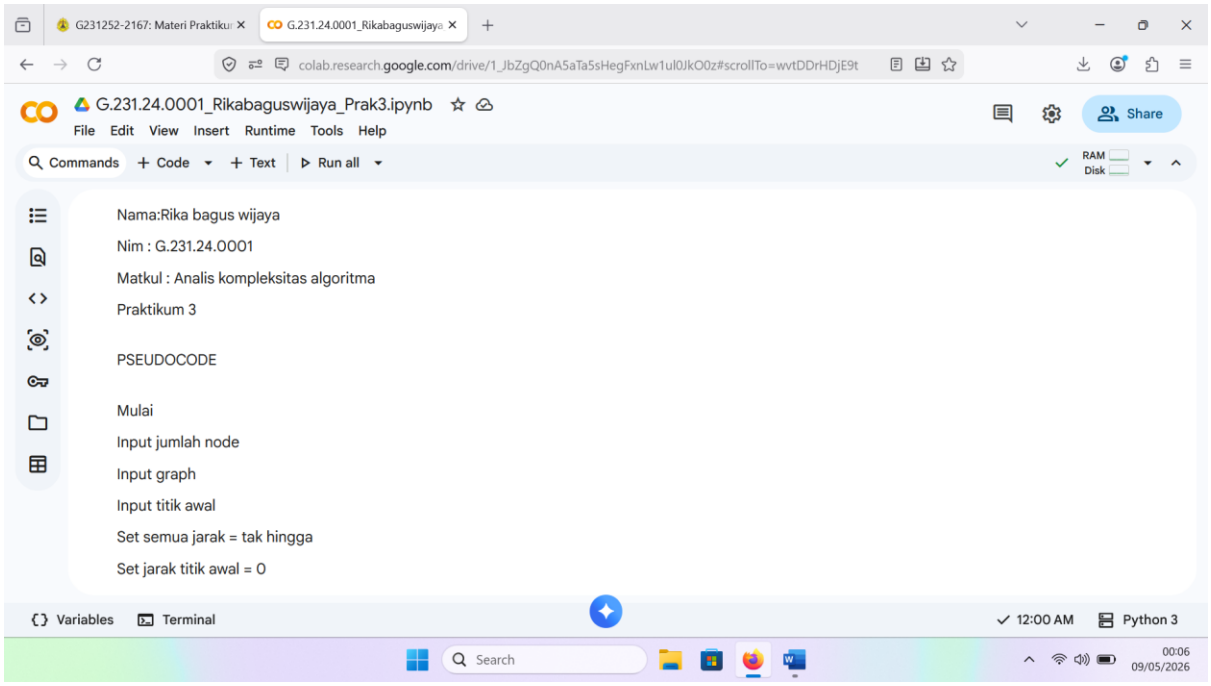
G231252-2167: Materi Praktiku
G.231.24.0001_Rikabaguswijaya
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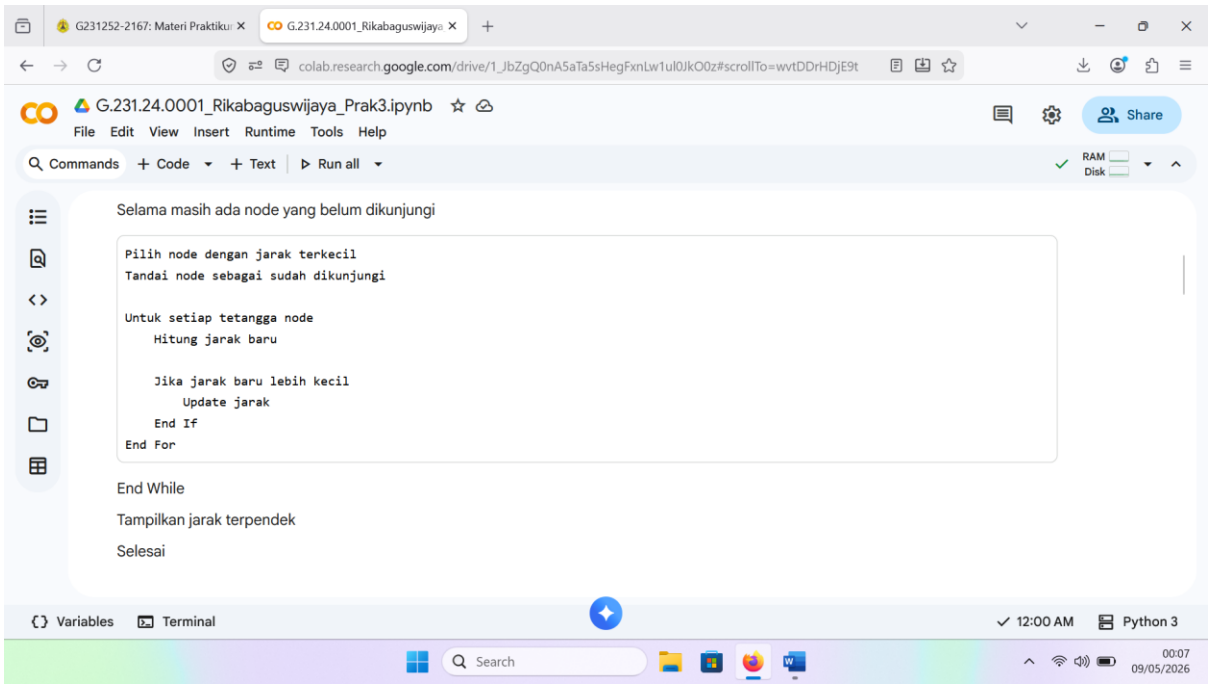
G.231.24.0001_Rikabaguswijaya_Prak3.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Nama:Rika bagus wijaya
Nim : G.231.24.0001
Matkul : Analis kompleksitas algoritma
Praktikum 3
PSEUDOCODE
Mulai
Input jumlah node
Input graph
Input titik awal
Set semua jarak = tak hingga
Set jarak titik awal = 0
Variables
Terminal
12:00 AM
Python 3
Search
00:06
09/05/2026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G.231.24.0001_Rikabaguswijaya_Prak3.ipynb
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RAM
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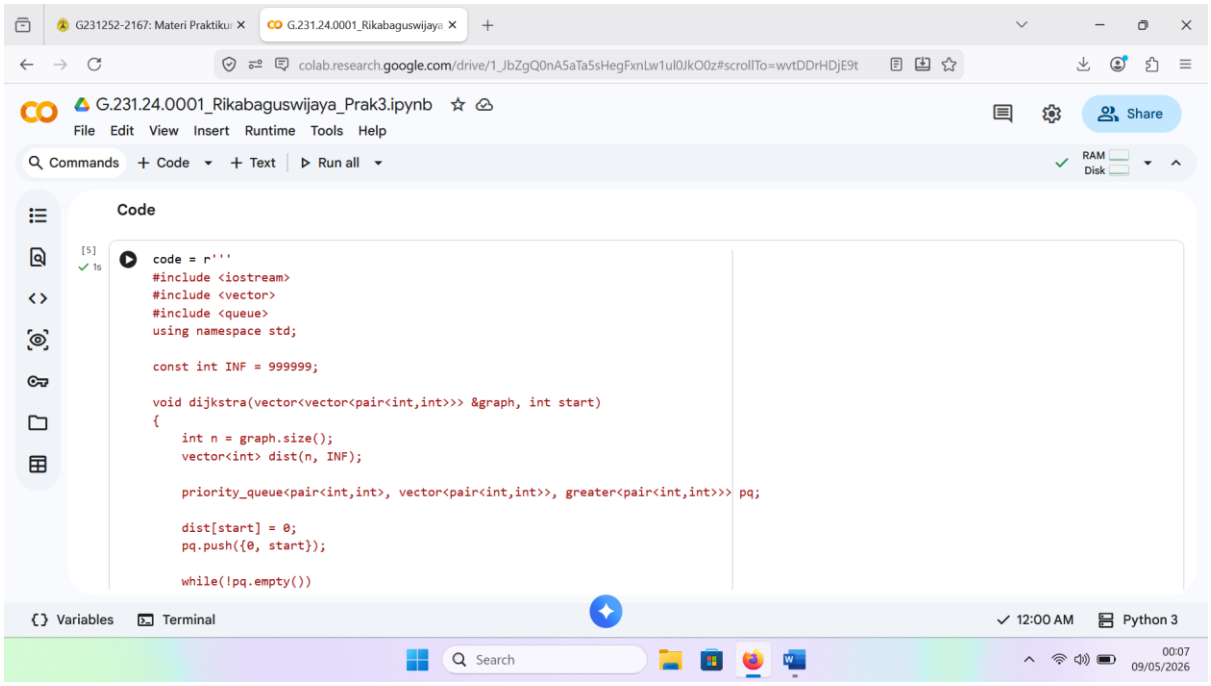
Selama masih ada node yang belum dikunjungi
Pilih node dengan jarak terkecil
Tandai node sebagai sudah dikunjungi
Untuk setiap tetangga node
Hitung jarak baru
Jika jarak baru lebih kecil
Update jarak
End If
End For
End While
Tampilkan jarak terpendek
Selesai
Variables
Terminal
12:00 AM
Python 3
Search
00:07
09/05/2026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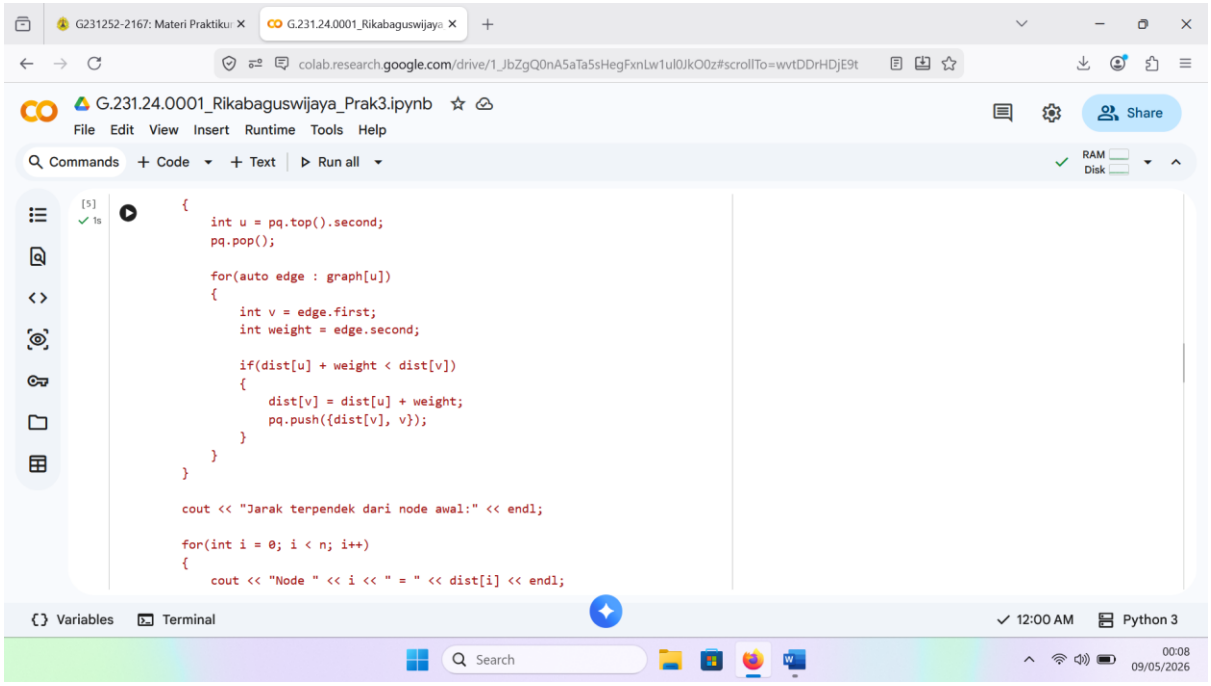
G.231.24.0001_Rikabaguswijaya_Prak3.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
Code
[5] 1s
code = r'''
#include <iostream>
#include <vector>
#include <queue>
using namespace std;
const int INF = 999999;
void dijkstra(vector<vector<pair<int,int>>> &graph, int start)
{
int n = graph.size();
vector<int> dist(n, INF);
priority_queue<pair<int,int>, vector<pair<int,int>>, greater<pair<int,int>>> pq;
dist[start] = 0;
pq.push({0, start});
while(!pq.empty())
Variables Terminal
12:00 AM
Python 3
Search
00:07
09/05/2026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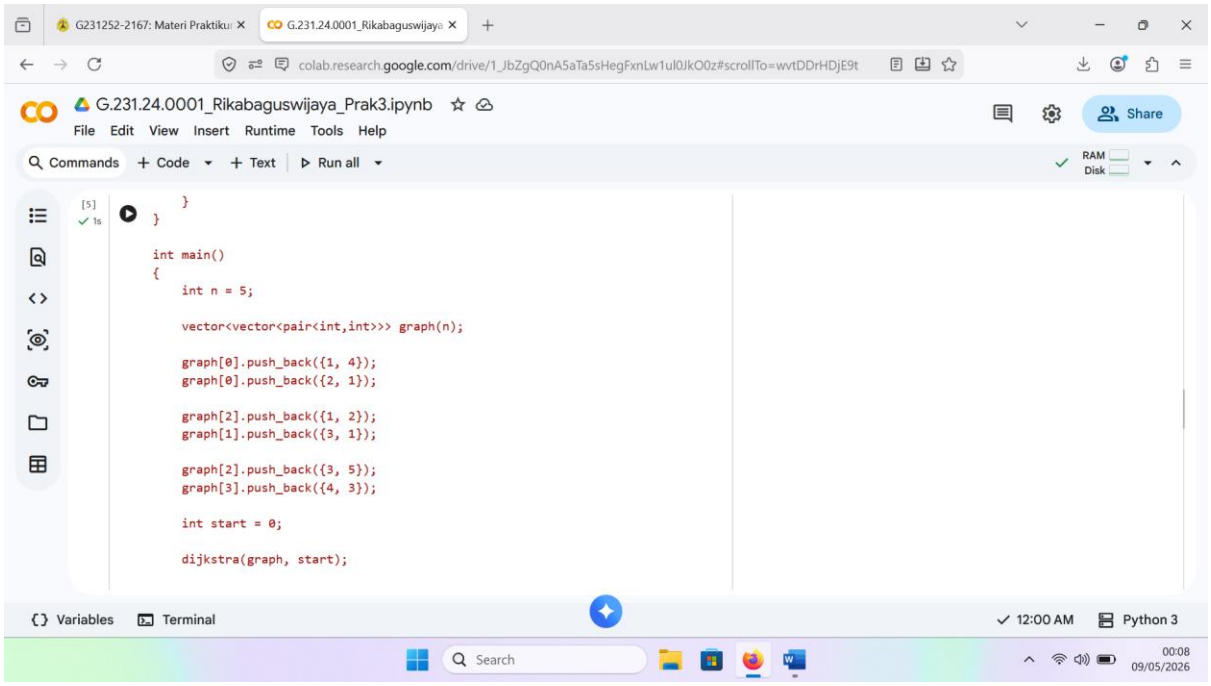
G.231.24.0001_Rikabaguswijaya_Prak3.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
[5] 1s
{
int u = pq.top().second;
pq.pop();
for(auto edge : graph[u])
{
int v = edge.first;
int weight = edge.second;
if(dist[u] + weight < dist[v])
{
dist[v] = dist[u] + weight;
pq.push({dist[v], v});
}
}
}
cout << "Jarak terpendek dari node awal:" << endl;
for(int i = 0; i < n; i++)
{
cout << "Node " << i << " = " << dist[i] << endl;
Variables Terminal
12:00 AM
Python 3
Search
00:08
09/05/2026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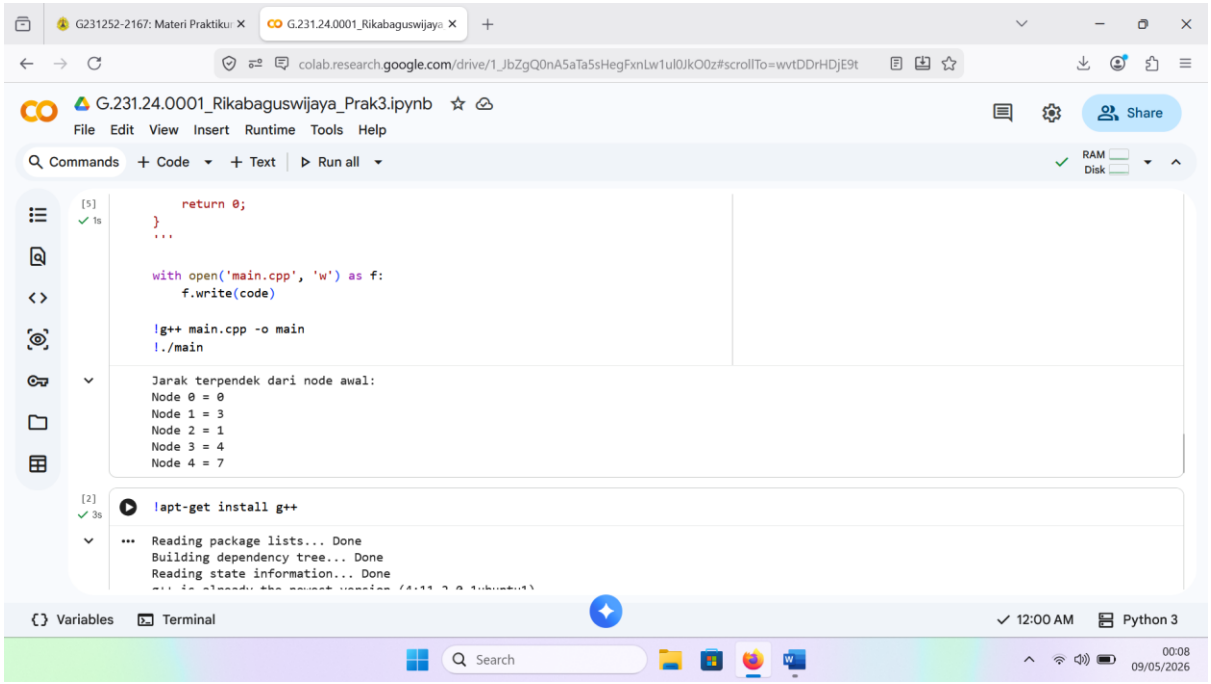
G.231.24.0001_Rikabaguswijaya_Prak3.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
[5]
1s
}
}
int main()
{
int n = 5;
vector<vector<pair<int,int>>> graph(n);
graph[0].push_back({1, 4});
graph[0].push_back({2, 1});
graph[2].push_back({1, 2});
graph[1].push_back({3, 1});
graph[2].push_back({3, 5});
graph[3].push_back({4, 3});
int start = 0;
dijkstra(graph, start);
Variables
Terminal
12:00 AM
Python 3
Search
00:08
09/05/2026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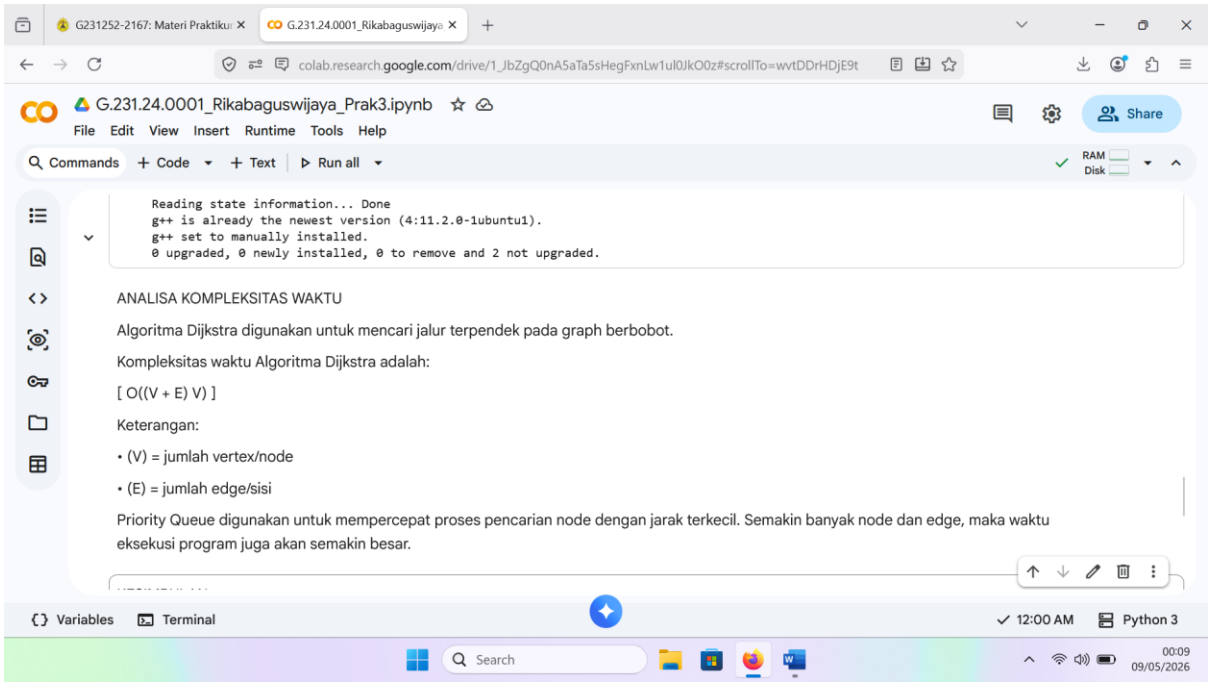
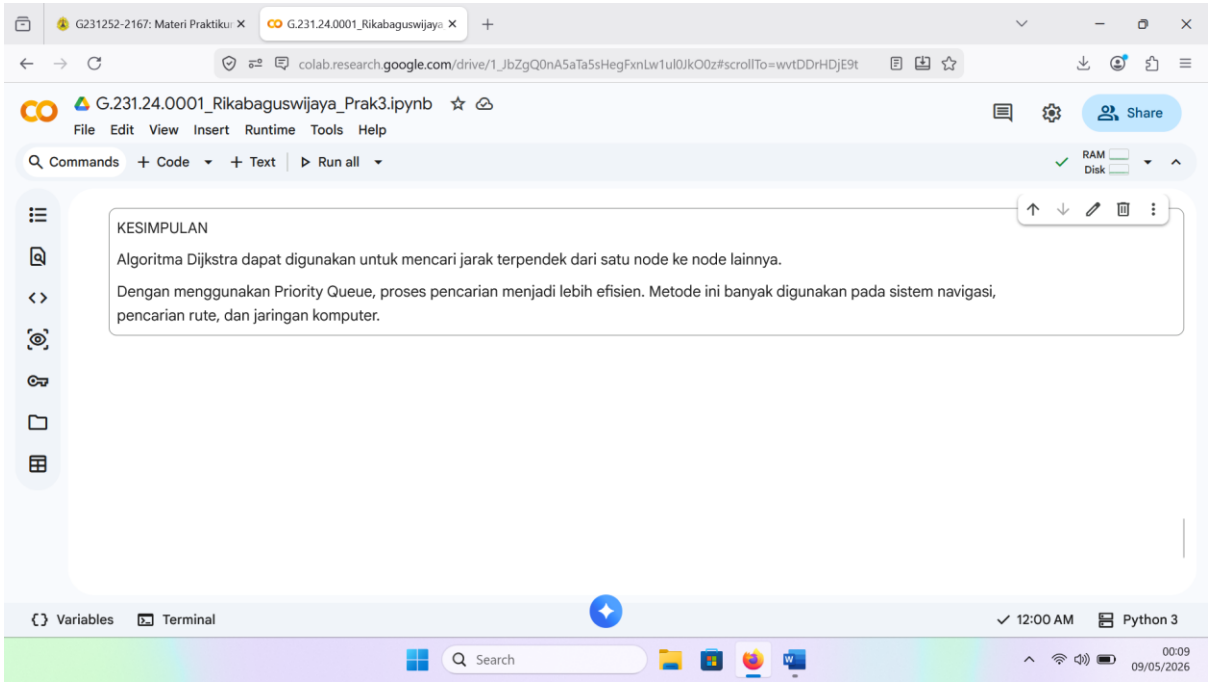
G.231.24.0001_Rikabaguswijaya
colab.research.google.com/drive/1_JbZgQ0nA5aTa5sHegFxnLw1ul0JkO0z#scrollTo=wvtDDrHDjE9t
G.231.24.0001_Rikabaguswijaya_Prak3.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
[5]
1s
return 0;
}
'''
with open('main.cpp', 'w') as f:
f.write(code)
!g++ main.cpp -o main
!./main
Jarak terpendek dari node awal:
Node 0 = 0
Node 1 = 3
Node 2 = 1
Node 3 = 4
Node 4 = 7
[2]
3s
!apt-get install g++
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Variables
Terminal
12:00 AM
Python 3
Search
00:08
09/05/2026





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